20. For a discussion of some of these proposals, see Tim Maudlin, *Quantum Nonlocalityand Relativity.*

Chapter 5

1. For the mathematically inclined reader, from the equation $t_{moving} = \gamma (t_{stationary} - (v/c^2) x_{stationary})$ (discussed in note 9 of Chapter 3) we find that Chewie's now-list at a given moment will contain events that observers on earth will claim happened $(v/c^2) x_{earth}$ earlier, where x_{earth} is Chewie's distance from earth. This assumes Chewie is moving away from earth. For motion toward earth, *v* has the opposite sign, so the earthbound observers will claim such events happened $(v/c^2)x_{earth}$ later. Setting v = 10 miles per hour and $x_{earth} = 10^{10}$ light-years, we find $(v/c^2) x_{earth}$ is about 150 years.

2. This number—and a similar number given in a few paragraphs further on describing Chewie's motion toward earth—were valid at the time of the book's publication. But as time goes by here on earth, they will be rendered slightly inaccurate.

3. The mathematically inclined reader should note that the metaphor of slicing the spacetime loaf at different angles is the usual concept of *spacetime diagrams* taught in courses on special relativity. In spacetime diagrams, all of three-dimensional space at a given moment of time, according to an observer who is considered stationary, is denoted by a horizontal line (or, in more elaborate diagrams, by a horizontal plane), while time is denoted by the vertical axis. (In our depiction, each "slice of bread"—a plane—represents all of space at one moment of time, while the axis running through the middle of the loaf, from crust to crust, is the time axis.) Spacetime diagrams provide an insightful way of illustrating the point being made about the now-slices of you and Chewie.